REV A LAYOUT REVIEW. NOT FOR FABRICATION.

Four–slot carrier: protected external 5 V / 4 A, isolated GPIO

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

U1
Pico 2 with headers, SC1632
J1
SSQ–110–01–G–S
H1
Captive M3 x 25 alignment post
H4
Captive M3 x 25 alignment post
R5
715

R20/R21: immediately after U5 SCK/MOSI outputs. R22: at U4 SO before shared MISO.
22R initial; 0R population option. Verify by oscilloscope/IBIS. No long stubs.
Each radio card has its own MISO source resistor before slot A7.

18/26 GPIO used; eight spare including GP26–28, all physically NC.
BUSY/IRQ/CS are direct GPIO through fixed–direction isolation, not expander GPIO.
KEY: captive M3 x 25 alignment post, see radio–slot–v0.1.md.
U4 GPA0..5 = S1/S3/S4 TX,RX pairs; GPB0..3 = S1..4 RESET_N.
No autonomous duty–cycled RX or automatic TX/RX transitions on Ebyte cards.

USB–only: Q1 off; RADIO_5V and IOREF_3V3 unpowered. External power: U2 –> radios and Q1 –> VSYS.
Q1 orientation is intentional: body diode from RADIO_5V toward VSYS. Pico onboard D1 blocks USB backfeed.
#FLG01/#FLG02 model the external supply; #FLG03 models real Q1/onboard USB–D1 ORing output.
U2 AUXOFF/FLT/ITIMER unused per datasheet. Leave ITIMER open for fastest current limiting.

SCHEMATIC REVIEW – NOT FOR FABRICATION

Sheet: /
File: carrier–rev–a.kicad_sch

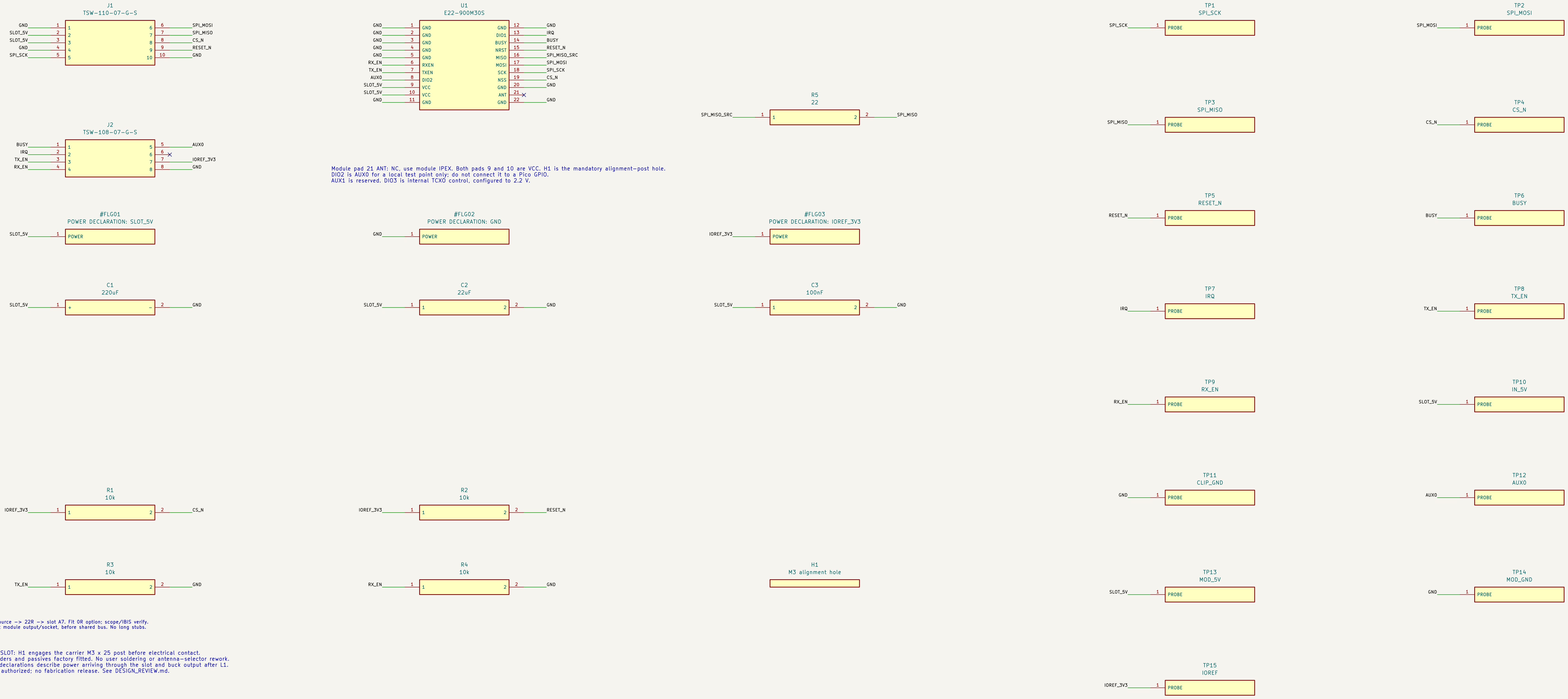
Title: Four–slot carrier: protected external 5 V / 4 A, isolated GPIO

Size: A0	Date: 2026–09–17	**Rev: A–schematic**
KiCad E.D.A. 10.0.6		Id: 1/1

REV A LAYOUT REVIEW. NOT FOR FABRICATION.

915 MHz LoRa card

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.


J1
TSW-110-07-G-S
GND 1, SLOT_5V 2, SLOT_5V 3, GND 4, SPI_SCK 5
6 SPI_MOSI, 7 SPI_MISO, 8 CS_N, 9 RESET_N, 10 GND
U1
E22-900M30S
GND 1 GND, GND 2 GND, GND 3 GND, GND 4 GND, GND 5 GND, RX_EN 6 RXEN, TX_EN 7 TXEN, AUX0 8 DIO2, SLOT_5V 9 VCC, SLOT_5V 10 VCC, GND 11 GND
GND 12 GND, DIO1 13 IRQ, BUSY 14 BUSY, NRST 15 RESET_N, MISO 16 SPI_MISO_SRC, MOSI 17 SPI_MOSI, SCK 18 SPI_SCK, NSS 19 CS_N, GND 20 GND, ANT 21, GND 22 GND
J2
TSW-108-07-G-S
BUSY 1, IRQ 2, TX_EN 3, RX_EN 4
5 AUX0, 6, 7 IOREF_3V3, 8 GND
R5
22
SPI_MISO_SRC 1 — 2 SPI_MISO
#FLG01
POWER DECLARATION: SLOT_5V
SLOT_5V 1 POWER
#FLG02
POWER DECLARATION: GND
GND 1 POWER
#FLG03
POWER DECLARATION: IOREF_3V3
IOREF_3V3 1 POWER
C1
220uF
SLOT_5V 1 + — 2 GND
C2
22uF
SLOT_5V 1 — 2 GND
C3
100nF
SLOT_5V 1 — 2 GND
R1
10k
IOREF_3V3 1 — 2 CS_N
R2
10k
IOREF_3V3 1 — 2 RESET_N
R3
10k
TX_EN 1 — 2 GND
R4
10k
RX_EN 1 — 2 GND
H1
M3 alignment hole
TP1 SPI_SCK — SPI_SCK 1 PROBE
TP2 SPI_MOSI — SPI_MOSI 1 PROBE
TP3 SPI_MISO — SPI_MISO 1 PROBE
TP4 CS_N — CS_N 1 PROBE
TP5 RESET_N — RESET_N 1 PROBE
TP6 BUSY — BUSY 1 PROBE
TP7 IRQ — IRQ 1 PROBE
TP8 TX_EN — TX_EN 1 PROBE
TP9 RX_EN — RX_EN 1 PROBE
TP10 IN_5V — SLOT_5V 1 PROBE
TP11 CLIP_GND — GND 1 PROBE
TP12 AUX0 — AUX0 1 PROBE
TP13 MOD_5V — SLOT_5V 1 PROBE
TP14 MOD_GND — GND 1 PROBE
TP15 IOREF — IOREF_3V3 1 PROBE

Module pad 21 ANT: NC, use module IPEX. Both pads 9 and 10 are VCC. H1 is the mandatory alignment-post hole.
DIO2 is AUX0 for a local test point only; do not connect it to a Pico GPIO.
AUX1 is reserved. DIO3 is internal TCXO control, configured to 2.2 V.

MISO: source -> 22R -> slot A7. Fit 0R option; scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

SCHEMATIC REVIEW – NOT FOR FABRICATION

Sheet: /
File: e22-900m30s-rev-a.kicad_sch

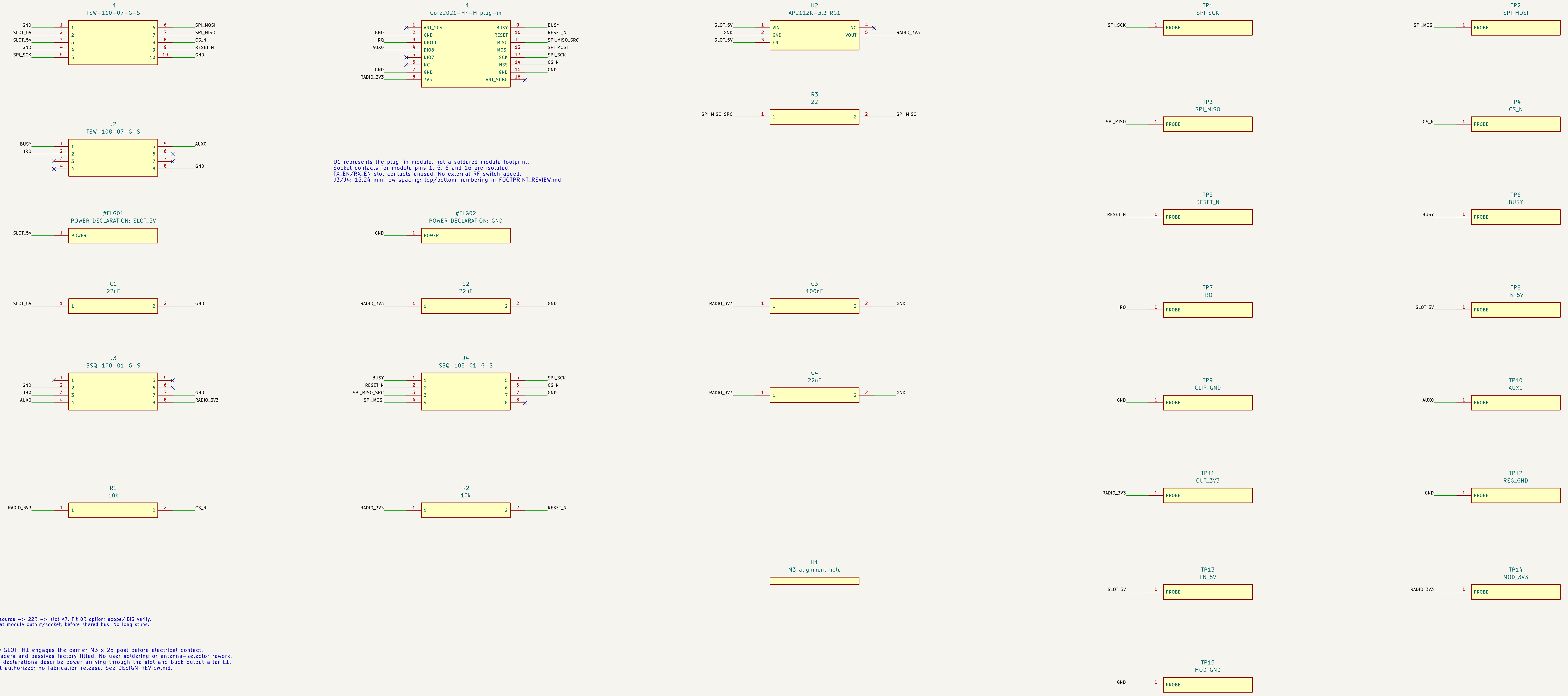
Title: 915 MHz LoRa card

Size: A2
Date: 2026-09-17
Rev: A-schematic
KiCad E.D.A. 10.0.6
Id: 1/1

REV A LAYOUT REVIEW. NOT FOR FABRICATION.

915 MHz FLRC socket adapter

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.


J1
TSW-110-07-G-S
GND 1, SLOT_5V 2, SLOT_5V 3, GND 4, SPI_SCK 5
6 SPI_MOSI, 7 SPI_MISO, 8 CS_N, 9 RESET_N, 10 GND
U1
Core2021-HF-M plug-in
1 ANT_2G4, GND 2 GND, IRQ 3 DIO11, AUX0 4 DIO8, 5 DIO7, 6 NC, GND 7 GND, RADIO_3V3 8 3V3
BUSY 9 BUSY, RESET 10 RESET_N, MISO 11 SPI_MISO_SRC, MOSI 12 SPI_MOSI, SCK 13 SPI_SCK, NSS 14 CS_N, GND 15 GND, ANT_SUBG 16
U2
AP2112K-3.3TRG1
SLOT_5V 1 VIN, GND 2 GND, SLOT_5V 3 EN
NC 4, VOUT 5 RADIO_3V3
R3
22
SPI_MISO_SRC 1, 2 SPI_MISO
J2
TSW-108-07-G-S
BUSY 1, IRQ 2, 3, 4
5 AUX0, 6, 7, 8 GND
#FLG01
POWER DECLARATION: SLOT_5V
SLOT_5V 1 POWER
#FLG02
POWER DECLARATION: GND
GND 1 POWER
C1
22uF
SLOT_5V 1, 2 GND
C2
22uF
RADIO_3V3 1, 2 GND
C3
100nF
RADIO_3V3 1, 2 GND
C4
22uF
RADIO_3V3 1, 2 GND
J3
SSQ-108-01-G-S
1, GND 2, IRQ 3, AUX0 4
5, 6, 7 GND, 8 RADIO_3V3
J4
SSQ-108-01-G-S
BUSY 1, RESET_N 2, SPI_MISO_SRC 3, SPI_MOSI 4
5 SPI_SCK, 6 CS_N, 7 GND, 8
R1
10k
RADIO_3V3 1, 2 CS_N
R2
10k
RADIO_3V3 1, 2 RESET_N
H1
M3 alignment hole
TP1 SPI_SCK — SPI_SCK 1 PROBE
TP2 SPI_MOSI — SPI_MOSI 1 PROBE
TP3 SPI_MISO — SPI_MISO 1 PROBE
TP4 CS_N — CS_N 1 PROBE
TP5 RESET_N — RESET_N 1 PROBE
TP6 BUSY — BUSY 1 PROBE
TP7 IRQ — IRQ 1 PROBE
TP8 IN_5V — SLOT_5V 1 PROBE
TP9 CLIP_GND — GND 1 PROBE
TP10 AUX0 — AUX0 1 PROBE
TP11 OUT_3V3 — RADIO_3V3 1 PROBE
TP12 REG_GND — GND 1 PROBE
TP13 EN_5V — SLOT_5V 1 PROBE
TP14 MOD_3V3 — RADIO_3V3 1 PROBE
TP15 MOD_GND — GND 1 PROBE

U1 represents the plug-in module, not a soldered module footprint.
Socket contacts for module pins 1, 5, 6 and 16 are isolated.
TX_EN/RX_EN slot contacts unused. No external RF switch added.
J3/J4: 15.24 mm row spacing; top/bottom numbering in FOOTPRINT_REVIEW.md.

MISO: source -> 22R -> slot A7. Fit 0R option; scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

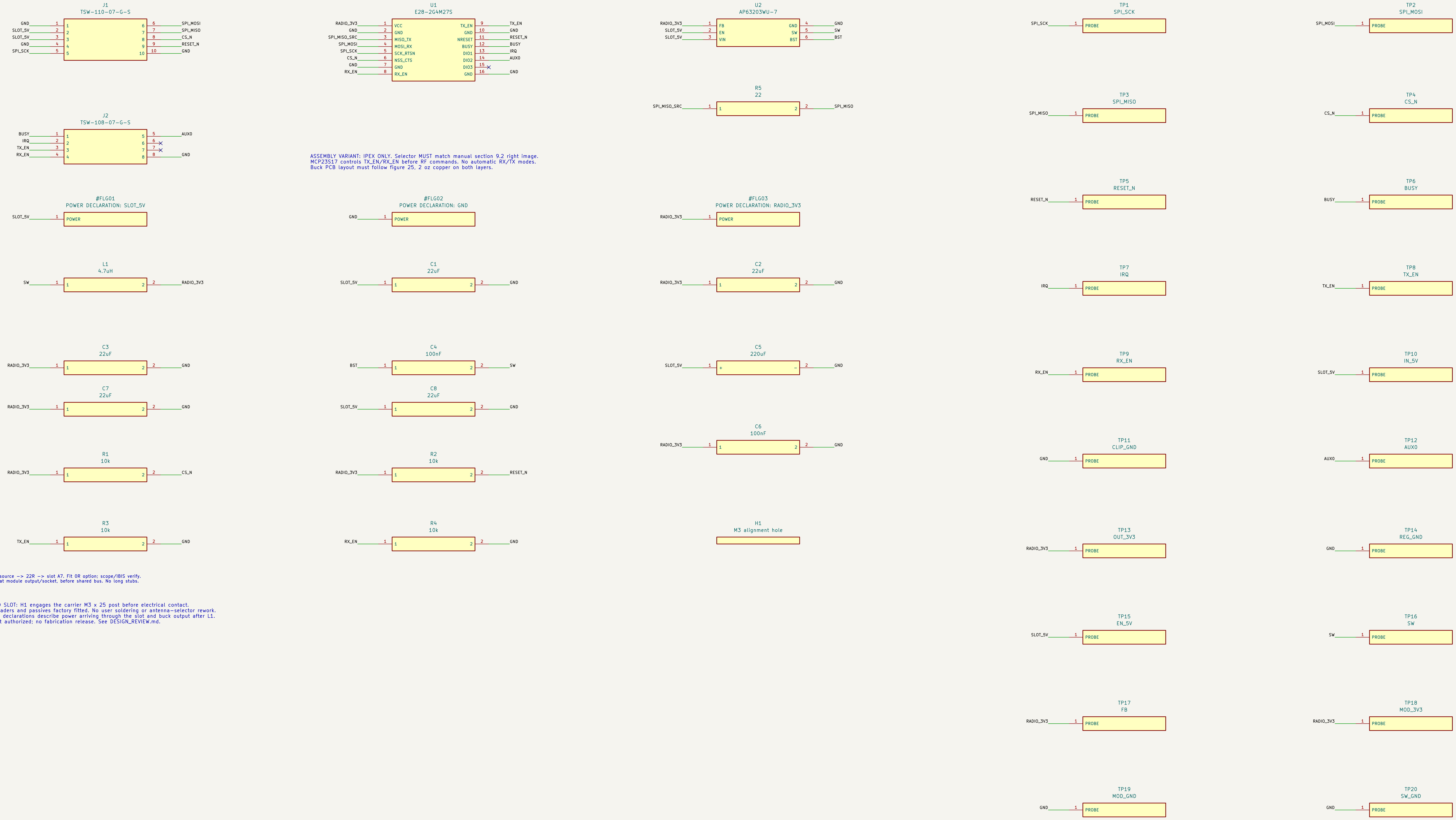
SCHEMATIC REVIEW — NOT FOR FABRICATION

Sheet: /
File: core2021-hf-m-rev-a.kicad_sch
Title: 915 MHz FLRC socket adapter
Size: A2
Date: 2026-09-17
Rev: A-schematic
KiCad E.D.A. 10.0.6
Id: 1/1

REV A LAYOUT REVIEW. NOT FOR FABRICATION.

2.4 GHz LoRa or FLRC card, same hardware

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.


J1
TSW-110-07-G-S
GND 1
SLOT_5V 2
SLOT_5V 3
GND 4
SPI_SCK 5
6 SPI_MOSI
7 SPI_MISO
8 CS_N
9 RESET_N
10 GND
U1
E28-2G4M27S
RADIO_3V3 1 VCC
GND 2 GND
SPI_MISO_SRC 3 MISO_TX
SPI_MOSI 4 MOSI_RX
SPI_SCK 5 SCK_RTSN
CS_N 6 NSS_CTS
GND 7 GND
RX_EN 8 RX_EN
TX_EN 9 TX_EN
GND 10 GND
NRESET 11 RESET_N
BUSY 12 BUSY
DIO1 13 IRQ
DIO2 14 AUX0
DIO3 15
GND 16 GND
U2
AP63203WU-7
RADIO_3V3 1 FB
SLOT_5V 2 EN
SLOT_5V 3 VIN
GND 4 GND
SW 5 SW
BST 6 BST
R5
22
SPI_MISO_SRC 1
2 SPI_MISO
J2
TSW-108-07-G-S
BUSY 1
IRQ 2
TX_EN 3
RX_EN 4
5 AUX0
6
7
8 GND
#FLG01
POWER DECLARATION: SLOT_5V
SLOT_5V 1 POWER
#FLG02
POWER DECLARATION: GND
GND 1 POWER
#FLG03
POWER DECLARATION: RADIO_3V3
RADIO_3V3 1 POWER
L1
4.7uH
SW 1 2 RADIO_3V3
C1
22uF
SLOT_5V 1 2 GND
C2
22uF
RADIO_3V3 1 2 GND
C3
22uF
RADIO_3V3 1 2 GND
C4
100nF
BST 1 2 SW
C5
220uF
SLOT_5V 1 + − 2 GND
C7
22uF
RADIO_3V3 1 2 GND
C8
22uF
SLOT_5V 1 2 GND
C6
100nF
RADIO_3V3 1 2 GND
R1
10k
RADIO_3V3 1 2 CS_N
R2
10k
RADIO_3V3 1 2 RESET_N
R3
10k
TX_EN 1 2 GND
R4
10k
RX_EN 1 2 GND
H1
M3 alignment hole
TP1
SPI_SCK
SPI_SCK 1 PROBE
TP2
SPI_MOSI
SPI_MOSI 1 PROBE
TP3
SPI_MISO
SPI_MISO 1 PROBE
TP4
CS_N
CS_N 1 PROBE
TP5
RESET_N
RESET_N 1 PROBE
TP6
BUSY
BUSY 1 PROBE
TP7
IRQ
IRQ 1 PROBE
TP8
TX_EN
TX_EN 1 PROBE
TP9
RX_EN
RX_EN 1 PROBE
TP10
IN_5V
SLOT_5V 1 PROBE
TP11
CLIP_GND
GND 1 PROBE
TP12
AUX0
AUX0 1 PROBE
TP13
OUT_3V3
RADIO_3V3 1 PROBE
TP14
REG_GND
GND 1 PROBE
TP15
EN_5V
SLOT_5V 1 PROBE
TP16
SW
SW 1 PROBE
TP17
FB
RADIO_3V3 1 PROBE
TP18
MOD_3V3
RADIO_3V3 1 PROBE
TP19
MOD_GND
GND 1 PROBE
TP20
SW_GND
GND 1 PROBE

ASSEMBLY VARIANT: IPEX ONLY. Selector MUST match manual section 9.2 right image.
MCP23S17 controls TX_EN/RX_EN before RF commands. No automatic RX/TX modes.
Buck PCB layout must follow figure 25, 2 oz copper on both layers.

MISO: source -> 22R -> slot A7. Fit 0R option; scope/IBIS verify.
Place at module output/socket, before shared bus. No long stubs.

KEYED SLOT: H1 engages the carrier M3 x 25 post before electrical contact.
All headers and passives factory fitted. No user soldering or antenna-selector rework.
Power declarations describe power arriving through the slot and buck output after L1.
Layout authorized; no fabrication release. See DESIGN_REVIEW.md.

SCHEMATIC REVIEW – NOT FOR FABRICATION

Sheet: /
File: e28-2g4m27s-rev-a.kicad_sch
Title: 2.4 GHz LoRa or FLRC card, same hardware
Size: A2 Date: 2026-09-17 **Rev: A-schematic**
KiCad E.D.A. 10.0.6 Id: 1/1